

# Open-Vocabulary 3D Scene Graphs from Sparse RGB Views: Duplicate-Aware Fusion and a De-Circularized Benchmark

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**Abstract.** We study how to build and—crucially—how to *honestly evaluate* open-vocabulary 3D scene graphs from a handful of casually captured RGB views. Our pipeline keeps every heavy model frozen and uses no per-scene optimization: feed-forward VGGT geometry [26] turns sparse images into world-point maps, an open-vocabulary detector (OWLv2) [15] proposes labeled object boxes, and these are lifted to 3D and fused across views into a labeled object–relation graph. Three findings follow once the labels are real and the evaluation is de-circularized. First, replacing the common class-agnostic-mask + per-patch CLIP front-end [10,20] with a genuine detector is what makes the labels real: on a desk scene it recovers the actual objects where mask+CLIP returned room-scale category noise. Second, the dominant failure of the lifted system is then *over-counting*—one physical object survives as several fused nodes—and a one-line duplicate-aware merge fixes it, improving object-label F1 at every view count and winning on 22–27 of the 28 benchmark scenes ( $p < 0.001$ ). Third, a previously promising uncertainty-aware fusion gate gives *no* benefit across a full weight sweep—a negative result that a circular evaluation had masked. To ground these claims we contribute an independent benchmark on 28 TUM RGB-D [23] sequences (5 human-verified, 23 two-pass VLM-drafted) whose reference is authored from the frames rather than seeded from the model’s own predictions, and we document the circular-pseudo-reference pitfall that inflates scores. The benchmark and de-circularization protocol are dataset-agnostic, and we release them for reuse.

**Keywords:** 3D scene graphs · sparse views · open-vocabulary detection · evaluation methodology · benchmark · duplicate-aware fusion

## 1 Introduction

Reconstructing a scene and explaining a scene are different goals. Feed-forward geometry models now recover camera motion, depth, and point maps directly from a few images [27,11,26], and 2D foundation models supply open-vocabulary detections, masks, and descriptors [15,20,10,17]. Yet downstream applications

sparse RGB views



open-vocabulary 3D scene graph



**Fig. 1.** From a handful of RGB views to an open-vocabulary 3D scene graph. **Left:** three of the input frames (TUM freiburg3\_long\_office\_household). **Right:** the system lifts frozen-model detections into VGGT world-point geometry and fuses them across views into labeled 3D object nodes (colored markers) — a desk, books, bottles, a cup, a bag, a box, and two office chairs — with all heavy models frozen and no per-scene optimization. Best viewed in color.

want object-level statements — which pixels across views are the same object, where it is in 3D, and how it relates to others (“the monitor is on the desk”) — and these components do not by themselves define such a graph, especially from a handful of casual RGB views rather than a calibrated RGB-D stream, dense video, or a prebuilt map.

We construct the graph as evidence-preserving fusion over lifted 2D proposals, with all heavy models frozen and no per-scene optimization: VGGT world-point maps act as a 2D-to-3D lookup, an open-vocabulary detector proposes labeled boxes, and these are lifted and fused across views into a labeled object-relation graph (Figure 1). This keeps perception fixed while making the graph stage explicit and easy to ablate, and targets a lower-input regime than open-vocabulary 3D graph systems built on posed RGB-D video or dense maps [4,6].

A central lesson of this work is methodological. An earlier version of this system graded its graphs against a pseudo-reference *seeded from the model’s own dense-view output* — a circular protocol under which a method that reproduces its dense graph scores 1.0 by construction. Re-grading against an *independently authored* reference, written from the frames, changes two conclusions. First, a mask-plus-CLIP front-end recovers almost none of the real objects (on a desk it returns “floor”/“curtain”/“bed” rather than the monitors and keyboard present); a genuine open-vocabulary *detector* (OWLv2) makes the labels real.

Second, a previously promising uncertainty-aware fusion gate shows *no* benefit once labels are real, across a full weight sweep — exactly the kind of conclusion a de-circularized evaluation is meant to surface. With honest labels, the dominant remaining error is *over-counting*: one physical object survives fusion as several same-label nodes (Figure 6). A one-line duplicate-aware merge of same-label nodes with overlapping 3D boxes fixes it, and is the only variant whose accuracy *rises* with more views — the opposite move from the uncertainty gate, which over-splits.

On 28 TUM RGB-D sequences [23] under 3/5/8/10 views, against an independent reference (5 human-verified, 23 two-pass drafted), duplicate-aware fusion improves object-label F1 over the graph-fusion baseline by +0.06 to +0.12 and wins on 22–27 of 28 scenes ( $p < 0.001$ ), while the uncertainty gate is consistently worse. Our contributions are:

- **A detector-grounded open-vocabulary sparse-view 3D scene-graph system** (OWLv2 + VGGT, all models frozen, no per-scene optimization), whose detector labels recover real objects where a mask-plus-CLIP front-end does not.
- **Duplicate-aware cross-view fusion**: a post-fusion merge of same-label nodes whose axis-aligned 3D boxes overlap, targeting the measured over-counting failure; it is the only variant whose accuracy rises with more views.
- **An independent, de-circularized benchmark on TUM RGB-D**: we document how a prediction-seeded reference inflates scores and show that an independently authored one removes the artifact.
- **A rigorous negative result on uncertainty-aware fusion**, with a mechanism analysis — it lifts recall but over-splits, costing precision — that also explains why merging works.

## 2 Related Work

*Feed-forward sparse-view geometry.* Classical multi-view reconstruction and SLAM rely on correspondence, triangulation, and bundle adjustment over calibrated or densely overlapping views (e.g. COLMAP [21], ORB-SLAM3 [2]). Feed-forward models remove the per-scene optimization: DUST3R predicts point maps from unconstrained images [27], MAST3R adds 3D-aware matching [11], and VGGT predicts cameras, depth, point maps, and confidence from one or many views [26]. We use VGGT as a *frozen* source of pixel-aligned 3D evidence, but our output is an object–relation graph, not a reconstruction.

*Open-vocabulary detection as a labeling front-end.* A lifted scene graph is only as good as its 2D labels. The common mask-then-embed recipe pairs class-agnostic SAM masks [10] with per-region CLIP embeddings [20] (optionally DINOv2 [17]); it is fully frozen but pushes naming onto whole-image-trained CLIP applied to crops, which tends to return scene-scale guesses rather than the objects present. Open-vocabulary *detectors* instead localize and name in one pass — OWL-ViT [16], OWLv2 [15], Grounding DINO [13], DetiC [28]. We adopt

OWLv2 because it emits localized, named instances — the property the lifting and fusion stages need — and study what that buys a sparse-view 3D pipeline over mask-then-embed.

*Open-vocabulary 3D semantics.* Open-vocabulary 3D understanding typically lifts or distills foundation-model features into point clouds, maps, or neural representations: building on NeRF [14], LERF embeds language in radiance fields [9], while OpenScene [18], ConceptFusion [6], OpenMask3D [24], and OpenSplat3D [19] fuse open-set features over points, maps, masks, or Gaussian splats [8]. We are complementary: we build explicit object-relation graphs from sparse RGB views without a completed 3D map or a per-scene radiance field. Notably, uncertainty-aware aggregation has been proposed for multi-view fusion in this setting [12]; we revisit it on real detector labels under an independent reference and report a negative result (Section 5).

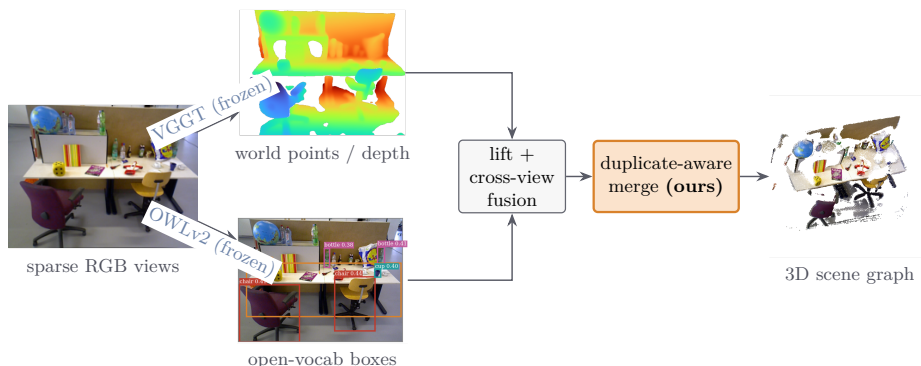
*3D scene graphs.* 3D scene graphs organize objects and their relations into structured scene representations [1], with datasets and learned predictors for reconstructed indoor scenes [25,3]. ConceptGraphs is closest to our output: it builds open-vocabulary 3D scene graphs from *posed RGB-D* sequences [4]. We share the object-relation goal but relax the input to sparse RGB views lifted through feed-forward geometry. A recurring failure of any such lifting-and-fusion pipeline is *over-counting* — one object surviving as several nodes — which motivates our duplicate-aware merge.

*Honest evaluation and reproducibility.* With no clean instance-level ground truth for a few casual RGB views, it is tempting to score a method against a reference derived from its own dense-view output — coupling reference to predictor and producing a *circular* evaluation in which a method scores perfectly at the reference view by construction. Such reference-predictor coupling is a form of information leakage, a known driver of the reproducibility crisis [7,5]. The 3D community has responded with controlled benchmarks carrying independent ground truth (e.g. Replica [22], TUM RGB-D [23]). In that spirit we contribute an independent sparse-view reference authored from the frames, document the circular-pseudo-reference pitfall, and report what actually helps once it is removed.

### 3 Method

#### 3.1 Problem Setup

Given a sparse set of RGB views  $\mathcal{I} = \{I_1, \dots, I_V\}$ , the goal is to produce an open-vocabulary 3D scene graph  $\mathcal{G} = (\mathcal{O}, \mathcal{R})$ . Each object node  $o_i \in \mathcal{O}$  stores a 3D point set, centroid, an axis-aligned 3D bounding box, an open-vocabulary label, view support, a feature descriptor, and its source detections. Each relation edge  $r_{ij} \in \mathcal{R}$  stores a spatial predicate between two object nodes. The method assumes neither depth images, externally supplied camera poses, nor a prebuilt



**Fig. 2.** The pipeline, on a real office scene. Sparse RGB views feed two *frozen* foundation models: VGGT produces world-point geometry (depth shown) and OWLv2 proposes open-vocabulary labeled boxes. Detections are lifted into the world-point map and fused across views into object nodes; our one addition is a *duplicate-aware merge* (amber) that collapses same-label overlapping nodes, yielding the labeled 3D object-relation graph at right. No model is fine-tuned and there is no per-scene optimization.

mesh: it derives geometry and labeled object proposals from frozen feed-forward models, lifts the proposals to a common 3D frame, and fuses them into objects with no per-scene optimization. Figure 2 summarizes the data flow.

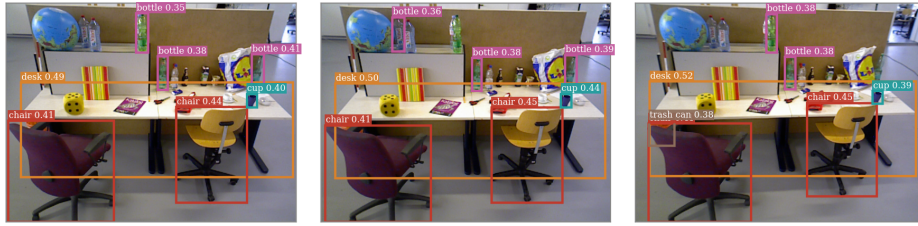
### 3.2 Overview

The pipeline has five frozen-model stages (Figure 2): VGGT geometry [26]; open-vocabulary detection with OWLv2, so each proposal already carries a real object label [15]; 3D lifting of each detection into the world-point map; cross-view fusion into object nodes, followed by the duplicate-aware merge; and rule-based spatial relations. The exported graph keeps predicted labels and relations alongside the evidence that produced them, so it can be inspected or turned into annotation packets.

Our one front-end change from the common open-vocabulary recipe is to replace mask-then-CLIP labeling (Section 2) with a genuine detector whose box *is* the labeled object hypothesis. Everything downstream is unchanged, so the resulting accuracy change is attributable to the front-end rather than to a re-tuned geometry stack.

### 3.3 Sparse-View Geometry

We run VGGT on the selected sparse views and store camera intrinsics, extrinsics, depth maps, world-point maps, and confidence maps [26]. The world-point map for each view acts as a 2D-to-3D lookup table: object proposals are lifted directly from image coordinates into a common 3D frame without first reconstructing a mesh, running bundle adjustment, or optimizing a radiance-field or



**Fig. 3.** The open-vocabulary detection front-end on real frames (three views of `freiburg3_long_office_household`). OWLv2, queried with the indoor vocabulary, returns named, boxed objects (desk, chair, bottle, cup, trash can) directly on each frame, with confidences — a genuine detector label per box. This is the property the lifting and fusion stages need; the common alternative (a post-hoc CLIP guess on a class-agnostic mask) returned room-scale category noise on these blurry frames rather than the objects actually present.

Gaussian-Splatting representation [21,14,8]. For view  $v$  we denote the world-point map by  $P_v(u) \in \mathbb{R}^3$  and the confidence map by  $C_v(u)$  at image coordinate  $u$ .

### 3.4 Open-Vocabulary Detection Front-End

For each view we run OWLv2 [15], an open-vocabulary detector that takes a set of free-form text queries and returns boxes with per-box class scores. The queries are the indoor object vocabulary of the benchmark, so the detector — not a post hoc patch-classification step — carries the open-vocabulary label. For detection  $k$  in view  $v$  we store its box, score  $s_{v,k}$ , and predicted label  $\ell_{v,k}$ . We use the `owlv2-base-patch16-ensemble` model with a detection threshold of 0.2 and per-class non-maximum suppression at box-IoU 0.5; the NMS mainly cleans the large surface classes (e.g. a desk) that otherwise accrue many overlapping boxes. We deliberately do *not* over-tighten the 2D threshold: a physical object may receive several overlapping boxes per frame, but overlapping boxes on the same object lift to nearly the same 3D points and are intended to collapse downstream. Optionally, each detection box prompts a frozen SAM [10] to produce a tight instance mask  $M_{v,k}$  so that lifting selects only object pixels; without a mask, lifting falls back to the box rectangle. When language-aligned and self-supervised appearance descriptors are available, we additionally extract CLIP and DINOv2 features for each detection and store the normalized concatenation [20,17]; these are used only for fusion gating and provenance, never for labeling.

### 3.5 3D Lifting

Each detection’s region (its instance mask  $M_{v,k}$ , or its box if no mask is available) is resized to the VGGT geometry resolution and used to select world points from the corresponding view; non-finite points and points below the confidence

threshold  $\tau_c$  are discarded:

$$X_{v,k} = \{P_v(u) \mid u \in M_{v,k}, C_v(u) \geq \tau_c\}. \quad (1)$$

Detections with no valid lifted points are rejected. Each surviving candidate node stores its sampled 3D point set  $X_{v,k}$ , centroid  $\mu_{v,k}$ , an axis-aligned 3D bounding box, the detector label  $\ell_{v,k}$  and score  $s_{v,k}$ , and (when available) the feature  $f_{v,k}$ .

### 3.6 Cross-View Fusion

Candidate nodes are fused into objects by building an association graph over the lifted detections. By default we disallow same-view association edges, so a connected component represents cross-view support for one object rather than a grouping of detections within a single image. An edge connects two candidates when their centroids are close in 3D and, when valid descriptors exist, their cosine similarity exceeds a feature threshold:

$$\|\mu_a - \mu_b\|_2 \leq \epsilon_g \quad \text{and} \quad \cos(f_a, f_b) \geq \epsilon_f. \quad (2)$$

The feature condition is omitted for candidates without valid descriptors. Connected components define fused object identities; for each component, points are pooled and sampled, descriptors are averaged and renormalized, and the node label is set by a score-weighted majority vote over the constituent detector labels  $\{\ell_{v,k}\}$  (ties broken by count). This separates view-local evidence from object identity while preserving the detection provenance needed for inspection and annotation.

### 3.7 Duplicate-Aware Fusion

Cross-view fusion alone leaves *over-counting* unaddressed, and two structural reasons make it expected: the detector over-fires (several overlapping boxes per object per frame), and the association graph forbids same-view edges by design, so duplicate detections of one object within a frame are never joined by cross-view fusion. A single desk or monitor therefore survives as several fused nodes — the dominant precision loss against the independent reference (Section 5). The corrective move is *merging*, not splitting.

We add a single post-fusion pass, DEDUP, that merges fused nodes of the same label whose axis-aligned 3D boxes overlap. Let  $\mathcal{O}$  be the fused nodes from Section 3.6. For a node  $o$  with point set  $X_o$ , let  $B_o = [b_o^{\min}, b_o^{\max}]$  be its axis-aligned 3D bounding box, with  $b_o^{\min} = \min X_o$  and  $b_o^{\max} = \max X_o$  taken coordinate-wise. For two boxes the axis-aligned 3D intersection-over-union is

$$\text{IoU}_{3D}(B_a, B_b) = \frac{\text{vol}(B_a \cap B_b)}{\text{vol}(B_a) + \text{vol}(B_b) - \text{vol}(B_a \cap B_b)}, \quad (3)$$

where  $\text{vol}(B_a \cap B_b) = \prod_{d=1}^3 \max(0, \min(b_{a,d}^{\max}, b_{b,d}^{\max}) - \max(b_{a,d}^{\min}, b_{b,d}^{\min}))$ . We partition  $\mathcal{O}$  by label. Within each label group we form a merge graph whose vertices

are the nodes and whose edges connect pairs with  $\text{IoU}_{3D}(B_a, B_b) > \tau$ , and we take its connected components; each component is collapsed into one object by re-running the same pooling, descriptor averaging, and label vote used in fusion. Nodes with no label, and singleton groups, are passed through unchanged. Concretely, with  $G_\ell = \{o \in \mathcal{O} : \text{label}(o) = \ell\}$ :

$$\text{DEDUP}(\mathcal{O}; \tau) = \bigcup_{\ell} \left\{ \text{FUSE}(c) : c \in \text{CC}(G_\ell, \{(a, b) : \text{IoU}_{3D}(B_a, B_b) > \tau\}) \right\}, \quad (4)$$

where  $\text{CC}(\cdot)$  denotes connected components and  $\text{FUSE}(\cdot)$  re-fuses a component into a single node (the identity on singletons). Restricting merges to same-label nodes keeps genuinely distinct objects of different classes separate, and requiring box overlap keeps spatially separated instances of the *same* class separate. The default threshold is  $\tau = 0.1$ ; the step is robust across the low-IoU range and the IoU criterion alone outperforms adding a centroid-distance term (Section 4). We use 3D-box IoU rather than mask or point-set overlap because the lifted boxes are exactly the geometry already attached to each node, making the test parameter-free beyond  $\tau$ . The one structural cost is that closely packed same-class instances (e.g. two monitors on one desk) can have overlapping boxes and be over-merged; we quantify this trade-off and discuss appearance-based instance splitting as the natural remedy in Sections 4 and 6.

### 3.8 Spatial Relations

Spatial relations are inferred from fused centroids using distance and relative-position rules. The relation vocabulary includes *near*, *left\_of*, *right\_of*, *above*, *below*, *in\_front\_of*, and *behind*. A *near* relation is emitted for object pairs within a distance threshold; otherwise the vertical displacement or the dominant horizontal coordinate determines the directional predicate, with an optional maximum relation distance suppressing very distant pairs. The exported graph preserves both the symbolic fields and the numeric support, so the same object identities, point sets, labels, and provenance can be reused by manual review or by a future learned relation head.

### 3.9 An Ablated Variant: Uncertainty-Aware Fusion

A natural hypothesis — and the headline of an earlier version of this system — is that fusion should be *uncertainty-aware*: each lifted candidate can be assigned an uncertainty

$$U_{v,k} = \lambda(1 - \bar{C}_{v,k}) + (1 - \lambda) \text{clip}\left(\frac{\sigma(X_{v,k})}{s_0}, 0, 1\right), \quad (5)$$

combining VGGT point confidence  $\bar{C}_{v,k}$  with the spatial spread  $\sigma(X_{v,k})$  of the lifted points (in our runs  $\lambda = 0.7$ ,  $s_0 = 1$ ), and the fusion gate is then *tightened* for uncertain pairs — effectively splitting ambiguous candidates apart. We retain this as an ablated variant rather than as part of the core method, because (i) the



**Table 1.** TUM RGB-D sequences prepared for sparse-view benchmarking. Each sequence contains 10 selected RGB frames and is evaluated with 3, 5, 8, and 10 input views.

| Scene ID                                 | Source sequence                 | Frames View counts |
|------------------------------------------|---------------------------------|--------------------|
| tum_rgbd_freiburg1_room                  | freiburg1_room                  | 10 3/5/8/10        |
| tum_rgbd_freiburg1_desk                  | freiburg1_desk                  | 10 3/5/8/10        |
| tum_rgbd_freiburg1_desk2                 | freiburg1_desk2                 | 10 3/5/8/10        |
| tum_rgbd_freiburg2_xyz                   | freiburg2_xyz                   | 10 3/5/8/10        |
| tum_rgbd_freiburg3_long_office_household | freiburg3_long_office_household | 10 3/5/8/10        |

failure analysis above shows the system over-counts, so the productive move is to merge more, not to split more, and a gate that splits uncertain pairs acts on the wrong axis; and (ii) once labels are real and the evaluation is de-circularized, the variant gives no benefit over plain cross-view fusion across a full weight sweep (Section 4). We report it as a deliberate, rigorous negative result. The uncertainty score itself is still exported with each node, since it is potentially useful for the opposite operation — guiding an appearance-based *split* of over-merged groups — which we leave to future work (Section 6).

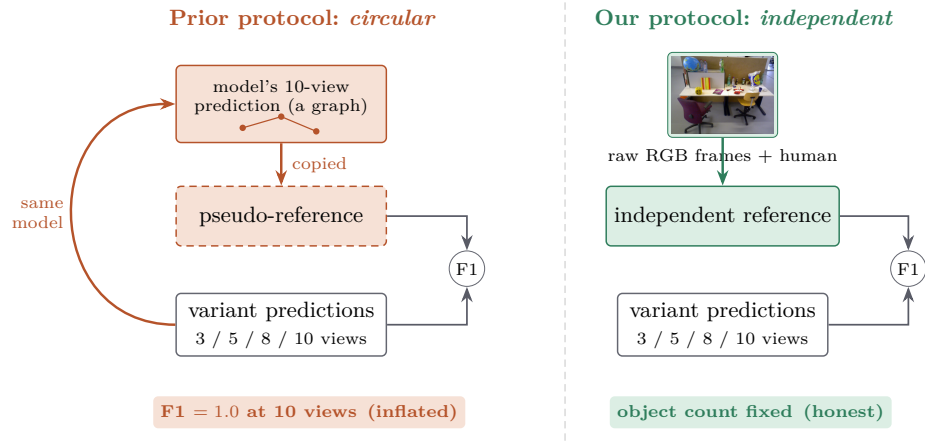
## 4 Experiments

We study two questions: what actually helps when building open-vocabulary 3D scene graphs from a handful of RGB views, and how to measure it honestly — the second matters because the system was originally reported as a sparse-view win for uncertainty-aware fusion, an artifact of a circular evaluation. We describe the benchmark (Sec. 4.1), its de-circularized reference (Sec. 4.2), and the variants and metric (Sec. 4.3–4.4) before the results (Section 5).

### 4.1 Benchmark Scenes and Sparse-View Protocol

We benchmark on public TUM RGB-D indoor sequences [23]. The benchmark spans **28 object-rich sequences** across **freiburg1–freiburg3**: the five paper-subset scenes in Table 1 (which carry the human-verified gold reference) plus 23 additional sequences; the full list is in `tum_rgbd_expanded.json`. Each sequence contributes ten selected RGB frames (stride 10). Although the source benchmark is RGB-D, our pipeline consumes only RGB; we use this dataset because it provides public, recognizable indoor scenes with reproducible sequence identifiers.

For each scene we evaluate  $N \in \{3, 5, 8, 10\}$  input views (a  $28 \times 4$  matrix, 112 graphs per variant), each run end-to-end along an identical code path — the pipeline of Section 3, driven by `run_owlv2_benchmark.sh`. Reporting accuracy as a function of  $N$  reads off sparse-view behavior directly rather than a single dense-view number.



**Fig. 4.** De-circularizing the evaluation. **Left (prior):** the reference is *copied* from the model’s own ten-view prediction, so the model is graded against itself: it scores 1.0 at ten views by construction, and a variant that splits more (the uncertainty gate) is rewarded for reproducing the reference’s fragments. **Right (ours):** the reference is authored from the raw RGB frames, independent of any model output, so the object count is fixed and over-splitting is penalized. Both findings — the uncertainty gate reversing to a loss, duplicate-aware fusion winning — appear only under the independent reference.

## 4.2 An Independently-Authored Reference

The central evaluation problem in this setting is that there is no ground-truth open-vocabulary object inventory for TUM RGB-D scenes, so a reference must be constructed. The tempting shortcut — and the one the original version of this work took — is to seed the reference from the system’s own dense-view prediction (the 10-view graph) and then lightly check it. We argue this is unsound and replace it with a reference authored *from the frames*.

*Why a prediction-seeded reference is circular.* Seeding the reference from a variant’s own 10-view output (Figure 4, left) makes that variant score 1.0 at ten views *by construction* and grades every sparser run on how faithfully it reproduces the same splitting pattern rather than agreement with the scene — so a method that keeps more proposals unmerged is rewarded for reproducing the reference’s own fragments. The quantitative artifacts this produced (the apparent uncertainty win, the dense-view ceiling) are in Section 5.6.

*Construction of the independent reference.* Our reference is, for each scene, an object multiset enumerated only from the ten raw RGB frames, independent of the detection/descriptor/geometry/fusion stack under test. Each scene is drafted once and re-examined in a second, adversarial pass that seeks missed or spurious objects. The five paper-subset scenes are additionally **human-verified** against

the frames (the gold core, `independent_labels.json`); the other 23 use the two-pass draft *without* verification (`independent_labels_expanded.json`) as scale corroboration, and both findings also hold on the gold five alone. Because the reference fixes the object inventory without reference to any model output, no variant can score 1.0 at ten views and over-splitting is penalized. It is a scene-level multiset over all ten frames, so sparse-view predictions are recall-limited against it — depressing absolute F1 uniformly across variants and leaving the comparison fair.

### 4.3 Variants Compared

We hold the front-end (OWLv2 detections lifted by VGGT geometry) and the reference fixed, and compare seven variants that differ only in how per-view detections become 3D nodes and how those nodes are fused across views:

- **2d-only** — per-view OWLv2 labels pooled without 3D lifting or cross-view fusion (an image-level upper reference on labels alone).
- **geometry-only** — detections lifted to 3D via VGGT and pooled, with no semantic cross-view association.
- **semantic-lifting** — detections lifted and associated by appearance descriptors without the geometric fusion gate.
- **fixed-shrink (uninformed control)** — the fusion gate is modulated by a fixed, scene-agnostic shrink factor; an uncertainty-blind control that isolates whether the *uncertainty signal*, rather than merely the act of tightening the gate, carries any benefit.
- **graph-fusion (baseline)** — the standard cross-view fusion of Section 3 with no uncertainty modulation and no duplicate handling.
- **proposed (uncertainty)** — graph-fusion with the rank-normalized uncertainty-aware merge gate (default weight  $w=0.3$ ); evaluated as an ablation rather than the headline method, and additionally swept over  $w \in \{0.1, 0.2, 0.3, 0.5, 0.8\}$ .
- **graph-fusion-dedup (ours)** — graph-fusion followed by a single post-fusion duplicate-instance merge: same-label fused nodes whose axis-aligned 3D boxes overlap ( $\text{IoU} > \tau$ , default  $\tau=0.1$ ) are unioned and re-fused (`merge_duplicate_instances` in `graph_builder.py`). This targets the measured over-counting failure (Section 5) and is the opposite move from the uncertainty gate, which *tightens* merging.

The pairing is deliberate: *fixed-shrink* versus *proposed* asks whether the uncertainty signal is informative beyond an uninformed gate adjustment, and *graph-fusion* versus *graph-fusion-dedup* asks whether merging *more* (rather than less) is the productive direction.

### 4.4 Metric

Our primary metric is `object_label_f1`: the multiset precision, recall, and F1 of fused-node labels against the independent reference multiset, averaged

over the 28 scenes (5 human-verified, 23 two-pass VLM-drafted). Treating labels as a multiset (rather than a set) makes the metric sensitive to over-counting — a scene with one physical monitor reported as four fused monitor nodes is penalized on precision — which is exactly the failure mode the reframed contribution targets. Per-scene wins/losses are reported alongside the means, with significance from a two-sided sign-test. Evaluation is run by `evaluate_sparse_view_annotations.py` and aggregated by `aggregate_variant_f1.py`; raw per-(scene,  $N$ , variant) scores are in `variant_independent_metrics.csv`.

#### 4.5 Implementation Details

All heavy models are frozen and run off-the-shelf; there is no per-scene optimization or fine-tuning. The open-vocabulary detector is OWLv2 (`google/owlv2-base-patch16-ensemble`) [15], the successor to OWL-ViT [16], with a detection score threshold of 0.2 and per-class non-maximum suppression at IoU 0.5; detection boxes prompt SAM [10] for masks. 3D geometry is the feed-forward VGGT transformer [26], consuming RGB frames only. Appearance descriptors for cross-view association use CLIP [20] and DINOv2 [17]. VGGT geometry runs on GPU; detection, lifting, fusion, and evaluation run on CPU. We fix all thresholds across scenes and view counts (no per-scene tuning), and the duplicate-aware merge uses a single global IoU threshold  $\tau=0.1$  that we show is robust across  $\tau \in [0, 0.2]$  (Section 5).

#### 4.6 Static and dynamic subsets

Six of the 28 sequences are *dynamic* (a moving person: `desk_with_person`, `sitting_*`, `walking_*`), which violates VGGT’s static multi-view assumption; the other 22 are static. We report the two as separate subsets and analyze the duplicate-aware gain on each with the results (Section 5.2).

### 5 Results

All numbers are object-label F1 against the independent 28-scene reference of Section 4 (multiset precision/recall/F1 of fused-node labels, mean over scenes at 3/5/8/10 input views). The contrast with the earlier prediction-seeded pseudo-reference is in Section 5.6.

#### 5.1 Variant comparison on the independent reference

Table 2 compares the seven variants of Section 4.3. On real, detector-grounded labels GRAPH-FUSION is the strongest prior fusion variant, yet object-label F1 *falls* as views are added for every cross-view fusion variant: more views supply more detections of the same physical object, which fusion does not collapse, so precision degrades against a fixed reference (the non-fusion references 2D-ONLY and GEOMETRY-ONLY merely plateau). GRAPH-FUSION-DEDUP is the only variant whose accuracy *rises* toward the dense end and is best at every view count (Figure 5).

**Table 2.** Object-label F1 on the independent reference (mean over 28 TUM RGB-D sequences; 5 human-verified + 23 two-pass VLM-drafted, full multiset). Best per column in bold. FIXED-SHRINK is an uninformed control; PROPOSED is the rank-normalized uncertainty-aware variant ( $w=0.3$ ). Accuracy falls with more views for every cross-view fusion variant except duplicate-aware fusion, which is best throughout.

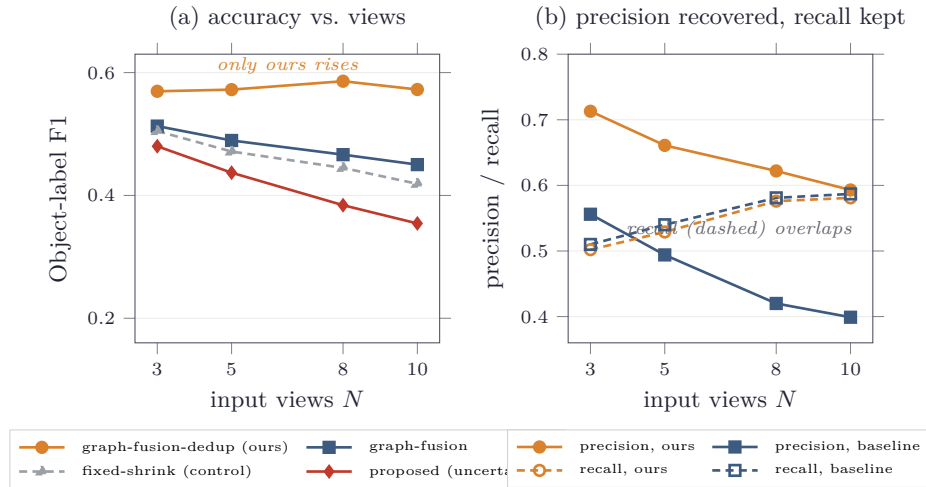
| Variant                          | v3            | v5            | v8            | v10           |
|----------------------------------|---------------|---------------|---------------|---------------|
| 2D-ONLY                          | 0.5120        | 0.5003        | 0.4867        | 0.4753        |
| GEOMETRY-ONLY                    | 0.4617        | 0.4605        | 0.4594        | 0.4493        |
| SEMANTIC-LIFTING                 | 0.3752        | 0.2881        | 0.2123        | 0.1797        |
| FIXED-SHRINK (control)           | 0.5048        | 0.4715        | 0.4448        | 0.4189        |
| GRAPH-FUSION (baseline)          | 0.5129        | 0.4895        | 0.4664        | 0.4502        |
| PROPOSED (uncertainty, $w=0.3$ ) | 0.4801        | 0.4370        | 0.3841        | 0.3544        |
| <b>graph-fusion-dedup (ours)</b> | <b>0.5696</b> | <b>0.5723</b> | <b>0.5860</b> | <b>0.5725</b> |

**Table 3.** Duplicate-aware fusion versus the GRAPH-FUSION baseline and the uninformed FIXED-SHRINK control.  $\Delta$  is the object-label F1 improvement; W/L/T counts are per-scene wins/losses/ties for dedup over GRAPH-FUSION ( $n = 28$ , sign test  $p < 0.001$  at every view).

|                                   | v3     | v5     | v8     | v10    |
|-----------------------------------|--------|--------|--------|--------|
| $\Delta$ dedup – GRAPH-FUSION     | +0.057 | +0.083 | +0.120 | +0.122 |
| $\Delta$ dedup – FIXED-SHRINK     | +0.065 | +0.101 | +0.141 | +0.154 |
| Per-scene W/L/T (vs GRAPH-FUSION) | 22/1/5 | 25/0/3 | 27/0/1 | 26/1/1 |

## 5.2 Duplicate-aware fusion: the positive result

Duplicate-aware fusion adds a single post-fusion step: same-label fused nodes whose axis-aligned 3D boxes overlap ( $\text{IoU} > \tau$ , default  $\tau=0.1$ ) are merged and re-fused. Table 3 reports the gain over both the GRAPH-FUSION baseline and the uninformed FIXED-SHRINK control. Dedup improves object-label F1 at every view count by +0.057 to +0.122 over GRAPH-FUSION and by +0.065 to +0.154 over FIXED-SHRINK; the gain *grows* with views, as expected for a step that removes the duplicate detections that accumulate as more frames are seen. It wins on 22–27 of 28 scenes at every view count (two-sided sign test  $p < 0.001$ ). The improvement holds on both the 22 static scenes (+0.082 to +0.171) and the 6 dynamic scenes with a moving person (+0.060 to +0.116), and on the 5 human-verified gold scenes alone (+0.068/ +0.089/ +0.145/ +0.146), so it is not an artifact of the VLM-drafted references.

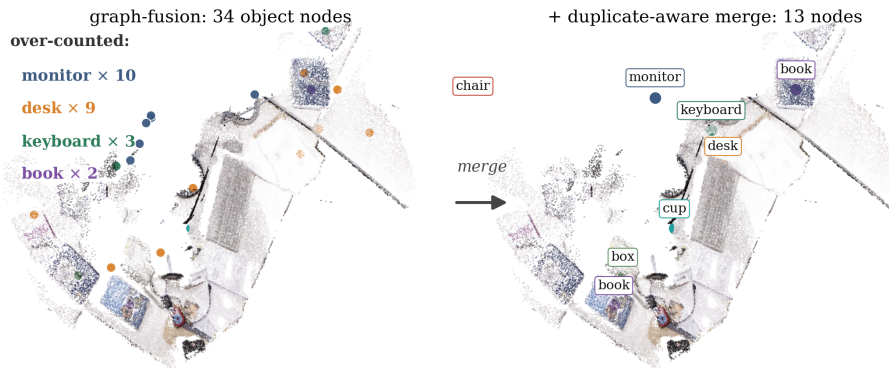


**Fig. 5.** Why duplicate-aware fusion wins, on the independent 28-scene reference. **(a)** Object-label F1 vs. number of input views: every prior cross-view fusion variant *degrades* as views are added (more frames yield more un-collapsed duplicates), while duplicate-aware fusion (amber) is the only variant whose accuracy *rises* and is best at every view count; the uncertainty gate (brick) sits below even the uninformed control. **(b)** The mechanism: dedup *recovers precision* (solid amber far above the baseline,  $0.71 \rightarrow 0.59$  vs.  $0.56 \rightarrow 0.40$ ) while leaving *recall essentially unchanged* (dashed lines overlap). Values match Table 2.

### 5.3 Uncertainty-aware fusion: a negative result

We treat the uncertainty-aware fusion gate — the headline of an earlier framing of this work — as an ablation. Table 4 reports its delta against GRAPH-FUSION. The recalibrated PROPOSED variant is *worse* at every view count ( $-0.033$  to  $-0.096$ ) and loses on essentially every scene (26–27 of 28 at v5–v10; sign test  $p < 0.001$ ). A sweep over the uncertainty weight  $w \in \{0.1, 0.2, 0.3, 0.5, 0.8\}$  does not rescue it: *no* weight in  $[0.1, 0.8]$  beats GRAPH-FUSION at any view count, with larger weights worse at the dense end. The result is robust to reference filtering: dropping **unknown object** (which the open-vocabulary detector never predicts) or the “stuff” classes (wall/floor/ceiling) raises every variant’s F1 but leaves the ranking unchanged.

The mechanism is informative rather than a mere null. The uncertainty modulation does exactly what it was designed to do — it lifts recall by declining to merge uncertain proposals (full-multiset recall, mean over scenes: PROPOSED  $0.516/0.556/0.596/0.602$  vs. GRAPH-FUSION  $0.510/0.540/0.581/0.587$ ) — but the same behavior fragments correct objects and costs more precision than the recall it buys (precision: PROPOSED  $0.478/0.388/0.299/0.266$  vs. GRAPH-FUSION  $0.556/0.495/0.420/0.399$ ). Against a fixed-count reference the net F1 is lower —



**Fig. 6.** Duplicate-aware fusion on the real `freiburg1_desk` point cloud (ten views; the cloud is desaturated so the object nodes stand out). **Left:** cross-view GRAPH-FUSION leaves the single monitor as ten nodes and the desk as nine — the over-counting that dominates the precision loss. **Right:** the duplicate-aware merge collapses same-label overlapping nodes into one object each (34  $\rightarrow$  13 nodes), recovering a clean labeled scene graph. (Merging is shown by 3D proximity for this illustration; the benchmark metric uses 3D-box IoU, Section 4.3.)

the *opposite* axis to duplicate-aware fusion, which merges *more* where the system over-splits (we develop this contrast in Section 6).

#### 5.4 Per-class characterization: the over-counting signature

Table 5 reports the micro-averaged precision/recall/F1 of GRAPH-FUSION per view count (dropping `unknown object`, which the detector never emits). The signature is clear: recall rises from 0.700 at three views to 0.802 at ten, while precision falls from 0.519 to 0.331, so net F1 is best at the sparse end (0.596 at v3, 0.468 at v10). The system finds more true objects with more frames but emits more duplicate detections, and the precision loss dominates.

Per class (at v8), this splits into three behaviors. A group is *found reliably but over-counted* — recall near 1.0 while precision collapses: `desk` (0.15 precision), `wall` (0.26), `floor` (0.27), `keyboard` (0.39), `monitor` (0.43), `chair` (0.45). These are always detected, but the 3D fusion does not collapse the several detections of one physical object, making them the dominant precision sink. A second group is *clean wins* with high precision and recall (`book`, `bottle`, `picture`, `person`, `cup`): distinct, well-separated objects that fuse correctly. A third group is *under-detected*: `cabinet` (recall 0.27), where the detector under-fires on blurry, low-resolution TUM frames. The dominant error is thus over-counting of correctly-found objects, not mislabeling — precisely what duplicate-aware fusion targets (Figure 7).

**Table 4.** Uncertainty-aware fusion is a negative result.  $\Delta$  PROPOSED  $-$  GRAPH-FUSION in object-label F1 (rank-normalized,  $w=0.3$ ); W/L/T is per-scene over  $n = 28$ . The variant is worse at every view count ( $p < 0.001$ ), and a weight sweep over  $[0.1, 0.8]$  contains no setting that beats the baseline.

|                                    | v3     | v5     | v8     | v10    |
|------------------------------------|--------|--------|--------|--------|
| $\Delta$ PROPOSED $-$ GRAPH-FUSION | -0.033 | -0.053 | -0.082 | -0.096 |
| Per-scene W/L/T                    | 2/22/4 | 0/26/2 | 0/26/2 | 0/27/1 |

**Table 5.** Per-class characterization of GRAPH-FUSION (micro-averaged over 28 scenes, dropping **unknown object**). Recall rises and precision falls as views are added; F1 is best at the sparse end. The precision loss is over-counting of correctly-found objects, not mislabeling.

| Views | Precision | Recall | F1           |
|-------|-----------|--------|--------------|
| 3     | 0.519     | 0.700  | <b>0.596</b> |
| 5     | 0.429     | 0.739  | 0.543        |
| 8     | 0.361     | 0.794  | 0.496        |
| 10    | 0.331     | 0.802  | 0.468        |

Table 6 confirms the mechanism at v8: dedup sharply raises precision on the over-counted classes (**desk**  $0.15 \rightarrow 0.45$ , **floor**  $0.27 \rightarrow 0.71$ , **monitor**  $0.43 \rightarrow 0.64$ , **cabinet**  $0.36 \rightarrow 0.67$ , **chair**  $0.45 \rightarrow 0.71$ ) while leaving recall unchanged for almost every class. The only recall cost is on closely-packed same-class instances — **desk** and **bottle** lose  $-0.04$  recall, as adjacent same-class objects with overlapping axis-aligned 3D boxes are merged. The net is a large F1 gain because the precision recovery dwarfs the recall cost; separating closely-packed same-class instances by appearance is left to future work (Section 6).

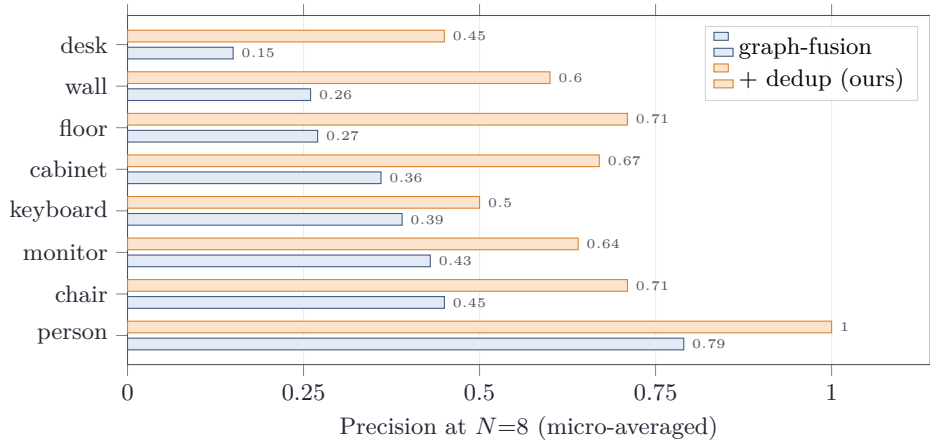
### 5.5 Robustness of the duplicate-aware merge

The dedup gain is robust to the IoU threshold (Table 7): the improvement over GRAPH-FUSION stays in the  $+0.075$  to  $+0.118$  range (average  $\Delta$ ) across the low-IoU band  $\tau \in [0, 0.2]$  and only fades once  $\tau$  is strict enough to stop merging duplicates ( $\tau \geq 0.5$ ,  $+0.030$ ). The default  $\tau=0.1$  is within noise of the optimum; a centroid-distance OR-term does not help, so 3D-box IoU alone is the right and simplest criterion.

### 5.6 De-circularization: the earlier pseudo-reference was misleading

The two findings above are visible only under an independent reference. The earlier evaluation seeded the reference from the model’s own ten-view GRAPH-FUSION output, producing two artifacts. (i) GRAPH-FUSION scored 1.0





**Fig. 7.** Where the gain comes from: per-class precision at eight views for the over-counted classes (Table 6). Duplicate-aware fusion sharply recovers precision on the classes the lifted system over-counts (**desk** 0.15  $\rightarrow$  0.45, **floor** 0.27  $\rightarrow$  0.71, **monitor** 0.43  $\rightarrow$  0.64) while leaving recall unchanged for almost every class. The error it fixes is over-counting of correctly-found objects, not mislabeling.

at ten views *by construction* — the reference *was* its own prediction. (ii) The uncertainty-aware variant, which keeps more proposals unmerged and so splits more, reproduced more of the fragments that reference contained and appeared to be a sparse-view *win* (+0.040/+0.021 at v3/v5). Fixing the true object count independently penalizes over-splitting: the uncertainty result reverses sign (Table 4) and the dense-view 1.0 artifact disappears (F1 falls with views, Table 5). A prediction-seeded reference is an easy, silent way to inflate scores in sparse-view 3D, and it masked both of this paper’s empirical findings.

## 6 Discussion

The paper’s findings are two sides of one mechanism. The lifted system’s dominant error is *over-counting*: one physical object survives cross-view fusion as several same-label nodes (Table 5). As views are added, recall rises but precision falls faster, so accuracy is best at the sparse end. Over-counting is an *over-splitting* error, so the corrective move is to *merge* — and the move in the opposite direction, an uncertainty gate that *tightens* merging, can only make it worse. This explains both empirical results at once.

### 6.1 Why merging helps and the uncertainty gate does not

Duplicate-aware fusion merges same-label nodes whose axis-aligned 3D boxes overlap, directly targeting the over-counting. It recovers precision on exactly the over-counted classes while preserving recall (Table 6), and is the only variant

**Table 6.** Per-class mechanism of duplicate-aware fusion at v8 (micro-averaged over 28 scenes). Dedup recovers precision on the over-counted classes while preserving recall for almost all classes; the only recall cost is on closely-packed same-class instances (**desk**, **bottle**).

| Class           | GRAPH-FUSION P/R | dedup P/R   | $\Delta$ recall |
|-----------------|------------------|-------------|-----------------|
| <b>monitor</b>  | 0.43 / 0.95      | 0.64 / 0.95 | 0.00            |
| <b>chair</b>    | 0.45 / 0.91      | 0.71 / 0.91 | 0.00            |
| <b>floor</b>    | 0.27 / 0.93      | 0.71 / 0.93 | 0.00            |
| <b>wall</b>     | 0.26 / 0.69      | 0.60 / 0.69 | 0.00            |
| <b>desk</b>     | 0.15 / 0.96      | 0.45 / 0.92 | −0.04           |
| <b>cabinet</b>  | 0.36 / 0.27      | 0.67 / 0.27 | 0.00            |
| <b>keyboard</b> | 0.39 / 1.00      | 0.50 / 1.00 | 0.00            |
| <b>person</b>   | 0.79 / 0.79      | 1.00 / 0.79 | 0.00            |
| <b>bottle</b>   | 0.70 / 0.76      | 0.78 / 0.72 | −0.04           |

whose accuracy *rises* with more views — each added view supplies redundancy the merge absorbs rather than fragmentation it cannot. The uncertainty-aware fusion gate acts on the same axis in the wrong direction: it tightens merging precisely where the system already over-splits. No weight in the sweep  $w \in [0.1, 0.8]$  beats the non-gated baseline, and the default regresses by  $-0.033$  to  $-0.096$  across the four view counts (Table 4). This is not a tuning failure but the predictable cost of suppressing merges in a regime whose error is too few merges; the contrast between the two variants is itself the clearest evidence for the over-counting diagnosis.

## 6.2 The circular-evaluation lesson

This mechanism was invisible under a prediction-seeded reference. When the reference is seeded from the model’s own dense-view output, a method that reproduces its ten-view graph scores 1.0 at the reference view *by construction*, and recall-raising behavior is rewarded whether or not it corresponds to real objects (Section 5.6); the uncertainty gate then looked like a sparse-view win. Re-grading the identical outputs against an independently authored reference reverses that result and exposes the precision collapse underneath. The takeaway generalizes: a reference seeded from a model measures agreement with that model, not recognition, and silently awards a perfect score to whatever produced the seed — an easy trap in sparse-view 3D, where dense ground truth is costly [5,7]. We therefore treat the independent reference and the documented pitfall as a first-class contribution, and recommend reporting any prediction-seeded reference alongside an independent one.

## 6.3 Limitations

*Reference provenance.* Of the 28 scenes, 5 are human-verified (the gold core) and 23 carry two-pass VLM-drafted references pending human verification. Both

**Table 7.** IoU-threshold sweep for duplicate-aware fusion (object-label F1, mean over 28 scenes). The gain over GRAPH-FUSION is robust across the low-IoU band  $[0, 0.2]$  and only fades for strict thresholds. Default  $\tau=0.1$  in bold.

| IoU $\tau$           | v3           | v5           | v8           | v10          | avg $\Delta$ |
|----------------------|--------------|--------------|--------------|--------------|--------------|
| 0.0 (any overlap)    | 0.581        | 0.593        | 0.610        | 0.605        | +0.118       |
| 0.05                 | 0.573        | 0.576        | 0.590        | 0.582        | +0.101       |
| <b>0.1 (default)</b> | <b>0.570</b> | <b>0.575</b> | <b>0.585</b> | <b>0.573</b> | +0.096       |
| 0.15                 | 0.565        | 0.568        | 0.573        | 0.561        | +0.087       |
| 0.2                  | 0.558        | 0.557        | 0.560        | 0.546        | +0.075       |
| 0.3                  | 0.544        | 0.535        | 0.534        | 0.522        | +0.054       |
| 0.5                  | 0.529        | 0.517        | 0.503        | 0.490        | +0.030       |

findings reach  $p < 0.001$  over the 28 scenes and hold on the gold five alone (Section 5.2), so they do not rest on the drafted labels; fully verifying the 23 is the natural next step, and the benchmark is small relative to large posed-RGB-D datasets.

*Closely-packed same-class instances.* Box-IoU merging cannot separate distinct same-class objects whose axis-aligned 3D boxes overlap, such as two monitors on one desk. This is the merge’s only recall cost —  $-0.04$  on **desk** and **bottle** at eight views, with other classes unchanged (Table 6) — and the precision gain dwarfs it.

*Static, “thing”-centric, rule-based.* The pipeline assumes a static scene, so VGGT’s multi-view geometry degrades on the dynamic (moving-person) sequences. “Stuff” classes (**wall**, **floor**, **ceiling**) are treated as countable objects, a poor fit for amorphous surfaces. And the spatial relations come from centroid/distance rules that are auditable but cannot express the support, containment, or functional relations of richer 3D scene graphs [1,25,4].

#### 6.4 Future work

The merge’s one failure mode points to the fix: appearance-based instance separation that splits a merged group when its members are visually distinct. This is also where the recall signal the uncertainty gate tried to add belongs — not as a merge gate that blocks fusion in the over-counting regime, but as a *split* decision on an already-merged group, the one operation where a recall signal helps rather than hurts. Replacing the rule-based predicates with a learned relation module, once the independent annotations supply enough supervision, is the other obvious extension.

## 7 Conclusion

We presented a detector-grounded open-vocabulary 3D scene-graph system that turns a handful of casual RGB views into an inspectable object–relation graph with all heavy models frozen and no per-scene optimization: OWLv2 [15] proposes labeled boxes, feed-forward VGGT [26] geometry lifts them to 3D, and cross-view fusion assembles the graph. On an independent TUM RGB-D [23] benchmark, the dominant error is over-counting, and a one-line duplicate-aware merge corrects it, improving object-label F1 by +0.06 to +0.12 at every view count and winning on 22–27 of the 28 scenes ( $p < 0.001$ ); a recalibrated, fully swept uncertainty-aware gate — the opposite move — gives no benefit, a negative result that a prediction-seeded reference had disguised as a win. The decircularized reference and the documented pitfall are the methodological core; the benchmark and protocol are dataset-agnostic, and we release them for reuse.

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